

LESSON 9-4 • SESSION 2 • PRACTICE • FORM B

Working Backward to a Goal

Learning target – I can work backward from a goal to find the input an equation needs. • 6.EE.C.9

REMEMBER

- **Work backward:** put the goal in the equation, then undo the math.
- With a fee, **subtract** it first, then **divide**: $20 = 4n + 8$, so $20 - 8 = 12$ and $12 \div 4 = 3$.
- If only whole items make sense, check whether you must **round up**.

WORKED EXAMPLE

Tickets cost \$6. A club needs at least \$100. How many tickets?

Step 1 Put in the goal: $100 = 6n$.

Step 2 Divide: $100 \div 6 \approx 16.7$.

Step 3 $16 \times 6 = \$96$ is too low; $17 \times 6 = \$102$ is enough.

Answer 17 tickets

WORDS **work backward** (*trabajar hacia atrás*) • **inverse operation** (*operación inversa*) • **goal** (*meta*)

PRACTICE

show your work • label units

1. Solve for x .

a. $8x = 96$ _____

b. $7x = 119$ _____

c. $3x + 6 = 45$ _____

d. $6x + 20 = 110$ _____

2. **Candle Sale.** A club sells candles for \$9 each. It needs to raise \$540.

a. Write an equation for the money m from p candles. _____

b. Put in the goal and work backward. _____ candles

c. If the club sells only 45 candles, what price must it charge? _____ dollars

3. **Walk-a-thon.** Sponsors pay Omar \$5 for each lap he walks. How many laps must he walk to raise \$185?

Hint Divide the goal by the pay for one lap.

Show your work.

Answer _____ laps

4. **Spring Play.** Tickets cost \$6 each. How many tickets must the club sell to raise \$324?

Hint Undo the multiplication.

Show your work.

Answer _____ tickets

5. **Charging Station.** A charger adds 7 miles of range every minute. Write an equation. How many minutes to add 168 miles?

Hint The goal replaces r . Then divide.

Show your work.

Answer _____ minutes

6. **Movie Tickets.** Tickets cost \$18 each. A group has \$200 and says it can buy 11 tickets. Does the plan work? What is the most it can buy?

Show your work.

Most tickets _____

7. **Band Trip Fund.** A band has \$80. It earns \$16 for each T-shirt sold. It needs \$560 in all.

Hint Subtract what the band already has first.

- a. Write an equation for the money m after k T-shirts are sold. _____
 b. How many T-shirts must be sold to reach \$560? _____

Show your work.

8. **Round Up.** A club earns \$9 for each ticket and needs \$300.

Hint You cannot sell part of a ticket.

- a. Solve $9t = 300$. _____
 b. How many tickets must the club sell? _____

FIND THE ERROR

explain the mistake, then fix it

9. **Darius's Work.** Stickers cost \$5 each. Darius needs \$85. He wrote $5s = 85$ and then answered 425 stickers. Explain his mistake and find the correct number.

TEST PRACTICE

choose the best answer

10. Solve $8n = 136$.

A 17 **B** 128 **C** 144 **D** 1,088

11. For $c = 6n + 12$, if $c = 48$, then n is –

A 10 **B** 6 **C** 300 **D** 8

SELF-CHECK CODE

find your answer · copy its letter into that problem's box · spell a math word

11 R	54 V	35 Q	60 R	37 E	925 X	24 E	33 Z	30 S	34 E
#2b		#3	#4	#5	#6	#7b	#8b		

CHECK YOURSELF ☐ I can do this on my own ☐ I need a little help ☐ I have a question about #____